

MENTORA ACADEMY

EXCELLENCE IN EDUCATION

Edexcel International A Level Chemistry (WCH14)

UNIT 4 — MASTER TOPICAL PRACTICE PACK: 15D

CARBOXYLIC ACID DERIVATIVES (ACYL CHLORIDES, ESTERS & POLYESTERS)

CANDIDATE NAME:	Usman
SUBJECT & CODE:	Chemistry Unit 4 — Rates, Equilibria & Organic (WCH14)
TOPIC AREA:	Topic 15: CARBOXYLIC ACID DERIVATIVES AND POLYESTERS
SUB-TOPIC:	15D: Carboxylic Acid Derivatives (Acyl Chlorides, Esters & Polyesters)
TOTAL QUESTIONS:	50 Authentic Past Paper Questions (Tier 1 & Tier 2)
TOTAL MARKS:	228 Marks
EXAMINER FAQs:	10 Empirical Examiner Traps & Model Answers Included

HOW TO USE THIS A* TOPICAL PACK:

1. Tier 1 (Q1–Q25): Mid-Easy to Moderate foundational questions (1–3 marks, MCQs, core calculations).
2. Tier 2 (Q26–Q50): Hard & A* Challenge questions (4–6 marks, multi-step calculations, multi-spectral NMR, extended answers).
3. Review the complete worked mark scheme in Section C after attempting each tier.
4. Master the 10 Empirical Examiner FAQs in Section D to eliminate common candidate traps.

SECTION A: TIER 1 PRACTICE QUESTIONS (FOUNDATIONAL TO MODERATE) — Q1 to Q25**Q1. 1. Ethanoyl Chloride Reactivity with Water****[4 Marks]****WCH14/01/Jan23/Q21**

Ethanoyl chloride (CH_3COCl) reacts vigorously with water at room temperature.

- (a) Write the balanced equation for the reaction of ethanoyl chloride with water. [1]

- (b) State the observation and name the gaseous product formed. [2]

- (c) Explain why ethanoyl chloride is much more reactive towards water than chlorobenzene. [1]

Q2. 2. Acid vs Alkaline Hydrolysis of Esters**[5 Marks]****WCH14/01/Oct22/Q22**

Compare the acid-catalysed hydrolysis and alkaline hydrolysis (saponification) of methyl propanoate.

- (a) Write the equation for the acid-catalysed hydrolysis of methyl propanoate using dilute H_2SO_4 . [2]

- (b) Write the equation for the alkaline hydrolysis of methyl propanoate using aqueous NaOH . [2]

- (c) Explain why alkaline hydrolysis goes to completion whereas acid hydrolysis is an equilibrium. [1]

Q3. 3. Synthesis of Acyl Chlorides using PCl_5 **[4 Marks]****WCH14/01/Jan22/Q14**

Propanoyl chloride is prepared from propanoic acid by reaction with phosphorus(V) chloride, PCl_5 .

(a) Write the balanced equation for this reaction. [1]

(b) State two observations that indicate the reaction has taken place. [2]

(c) State why this reaction must be carried out under dry conditions. [1]

Q4. 4. Acyl Chloride Reaction with Ammonia

[4 Marks]

WCH14/01/June21/Q18

Ethanoyl chloride reacts with concentrated aqueous ammonia to form ethanamide.

(a) Write the balanced equation for this reaction. [2]

(b) State why two moles of ammonia are required per mole of ethanoyl chloride. [1]

(c) Identify the functional group present in ethanamide. [1]

Q5. 5. Preparation of N-Ethylpropanamide

[4 Marks]

WCH14/01/Jan21/Q19

Propanoyl chloride reacts with ethylamine ($\text{CH}_3\text{CH}_2\text{NH}_2$) to yield an N-substituted amide.

(a) Draw the structural formula of N-ethylpropanamide. [1]

(b) Write the equation for this reaction. [2]

(c) Classify the mechanism of this reaction. [1]

Q6. 6. Esterification of Benzoyl Chloride with Ethanol

[3 Marks]

WCH14/01/Oct20/Q15

Benzoyl chloride ($\text{C}_6\text{H}_5\text{COCl}$) is used to prepare ethyl benzoate.

(a) Write the balanced equation for the reaction of $\text{C}_6\text{H}_5\text{COCl}$ with ethanol. [1]

(b) Give two advantages of using benzoyl chloride over benzoic acid to synthesize ethyl benzoate. [2]

Q7. 7. Ester Naming and Isomerism

[4 Marks]

WCH14/01/Jan20/Q08

An ester with molecular formula $\text{C}_4\text{H}_8\text{O}_2$ is hydrolysed to produce ethanoic acid and ethanol.

(a) Name this ester and draw its displayed formula. [2]

(b) Draw the structural formula of a structural isomer of $\text{C}_4\text{H}_8\text{O}_2$ that is also an ester and yields methanol on hydrolysis. [1]

(c) State the IUPAC name of the ester drawn in (b). [1]

Q8. 8. Hydrolysis of Vegetable Oils (Triglycerides)**[5 Marks]****WCH14/01/June19/Q20**

Triglycerides present in vegetable oils undergo alkaline hydrolysis with aqueous NaOH to produce soap.

(a) Name the alcohol formed as a byproduct during saponification of triglycerides. [1]

(b) Draw the structural formula of this alcohol. [1]

(c) Write an equation for the saponification of a triglyceride containing three glyceryl trioctadecanoate ester links. [3]

Q9. 9. Acylating Agents Comparison: Acyl Chloride vs Acid Anhydride**[4 Marks]****WCH14/01/Jan19/Q12**

Aspirin (acetylsalicylic acid) can be synthesized by acylating 2-hydroxybenzoic acid using either ethanoyl chloride or ethanoic anhydride.

(a) Draw the structural formula of ethanoic anhydride. [1]

(b) State two reasons why ethanoic anhydride is preferred over ethanoyl chloride in industrial synthesis of aspirin. [2]

(c) Name the organic byproduct formed when ethanoic anhydride is used. [1]

Q10. 10. Polyester Formation: Monomers of Terylene**[4 Marks]**
WCH14/01/June18/Q17

PET (Terylene) is manufactured by reacting benzene-1,4-dicarboxylic acid with ethane-1,2-diol.

- (a) State the type of polymerisation involved in forming PET. [1]

- (b) Draw the structural formulas of both monomer molecules. [2]

- (c) Identify the small molecule eliminated during polymerisation. [1]

Q11. 11. Repeat Unit of Polylactic Acid (PLA)**[3 Marks]**
WCH14/01/Jan18/Q16

Polylactic acid (PLA) is a biodegradable polymer synthesized from 2-hydroxypropanoic acid (lactic acid).

- (a) Explain why lactic acid can polymerise with itself to form a polyester. [1]

- (b) Draw the repeat unit of PLA showing continuation bonds. [2]

- (c) State why PLA is described as a biodegradable polymer. [1]

Q12. 12. Reactions of Methyl Methanoate**[3 Marks]**
WCH14/01/Oct17/Q13

Methyl methanoate (HCOOCH_3) is the simplest ester.

- (a) State the expected observation when methyl methanoate is heated with Fehling's solution. [2]

(b) Explain why methyl methanoate gives a positive result with Fehling's solution whereas methyl ethanoate does not. [1]

Q13. 13. Physical Properties of Esters vs Carboxylic Acids

[3 Marks]
WCH14/01/June17/Q11

Ethyl ethanoate and butyric acid (butanoic acid) are structural isomers with molecular formula $C_4H_8O_2$.

(a) State which compound has the higher boiling point. [1]

(b) Explain your answer in terms of intermolecular forces. [2]

Q14. 14. Reaction of Acyl Chloride with Phenol

[4 Marks]
WCH14/01/Jan17/Q15

Phenyl ethanoate is produced by reacting phenol (C_6H_5OH) with ethanoyl chloride in the presence of base.

(a) Write the equation for this reaction. [2]

(b) Explain why phenol does not react directly with ethanoic acid to form an ester. [2]

Q15. 15. Synthesis of Paracetamol

[4 Marks]
WCH14/01/Oct16/Q18

Paracetamol (N-(4-hydroxyphenyl)ethanamide) is prepared by reacting 4-aminophenol with ethanoic anhydride.

(a) Draw the structural formula of paracetamol. [1]

(b) Explain why the amine group (-NH_2) of 4-aminophenol reacts preferentially over the phenol (-OH) group. [2]

(c) Write the balanced equation for the reaction. [1]

Q16. 16. Polyester Kodel Synthesis

[4 Marks]

WCH14/01/June16/Q14

The polyester Kodel is formed from 1,4-bis(hydroxymethyl)cyclohexane and dimethyl terephthalate.

(a) State the type of reaction involved when an ester reacts with a diol to form a polyester. [1]

(b) Name the small molecule eliminated in this transesterification polymerisation. [1]

(c) Draw the ester linkage present in Kodel. [2]

Q17. 17. Acid Hydrolysis of Polyesters

[4 Marks]

WCH14/01/Jan16/Q19

A sample of PET is refluxed with concentrated hydrochloric acid.

(a) State the names of the two organic products formed upon complete hydrolysis. [2]

(b) Write equations showing the hydrolysis of one ester link in PET. [2]

Q18. 18. Alkaline Hydrolysis of Polyesters

[4 Marks]
WCH14/01/Oct15/Q12

PET is treated with hot aqueous sodium hydroxide solution.

(a) Draw the structural formulas of the two organic products formed. [2]

(b) Explain why the dicarboxylic acid is obtained as a salt rather than the free acid. [2]

Q19. 19. Preparation of Acyl Chloride using SOCl_2

[3 Marks]
WCH14/01/June15/Q16

Thionyl chloride (SOCl_2) reacts with ethanoic acid to produce ethanoyl chloride.

(a) Write the balanced equation for this reaction. [1]

(b) State one operational advantage of using SOCl_2 over PCl_5 in this synthesis. [2]

Q20. 20. Reactivity Trend of Carbonyl Derivatives

[4 Marks]
WCH14/01/Jan15/Q10

Consider the relative rates of hydrolysis for acyl chlorides, acid anhydrides, esters, and amides.

- (a) Arrange these four carboxylic acid derivatives in order of decreasing reactivity towards nucleophilic attack. [1]

- (b) Explain why acyl chlorides are the most reactive towards nucleophiles. [3]

Q21. 21. Esterification Catalyst Function

[3 Marks]
WCH14/01/Oct14/Q14

Concentrated sulfuric acid is added as a catalyst in esterification reactions.

- (a) Describe the catalytic role of H^+ ions in the mechanism of esterification. [2]

- (b) State a secondary function of concentrated H_2SO_4 in driving ester yield. [1]

Q22. 22. Hydrolysis Rate of Acyl Chlorides vs Aliphatic Halides

[4 Marks]
WCH14/01/June14/Q15

Compare the rate of reaction of water with ethanoyl chloride and 1-chloropropane.

- (a) State which compound hydrolyses faster. [1]

- (b) Explain the difference in mechanism and activation energy. [3]

Q23. 23. Ester Hydrolysis in Acidic Buffer**[4 Marks]**
WCH14/01/Jan14/Q11

An ester is hydrolysed in an aqueous solution buffered at pH 3.0.

- (a) State the chemical forms of the products obtained at pH 3.0. [2]

- (b) Explain how changing the pH to 12.0 affects the position of equilibrium and products. [2]

Q24. 24. Condensation Polymer Polycaprolactone**[4 Marks]**
WCH14/01/Oct13/Q17

Polycaprolactone (PCL) is a biodegradable polyester formed from 6-hydroxyhexanoic acid.

- (a) Write the formula of 6-hydroxyhexanoic acid. [1]

- (b) Draw a segment of polycaprolactone containing two repeat units. [2]

- (c) Suggest an biomedical application of polycaprolactone based on its biodegradability. [1]

Q25. 25. Transesterification in Biodiesel Production**[4 Marks]**
WCH14/01/June13/Q19

Biodiesel consists of methyl esters of long-chain fatty acids produced by transesterification of vegetable oil with methanol in the presence of KOH catalyst.

- (a) Write a general equation for transesterification of a triester with methanol. [2]

(b) Explain why vegetable oil cannot be used directly in diesel engines without conversion to biodiesel. [2]

SECTION B: TIER 2 A* CHALLENGE QUESTIONS (HARD & EXTENDED) — Q26 to Q50

Q26. 26. Nucleophilic Addition-Elimination Mechanism of Acyl Chlorides

[6 Marks]

WCH14/01/Jan23/Q25

Ethanoyl chloride reacts with methanol to form methyl ethanoate and hydrogen chloride.

(a) Draw the full step-by-step nucleophilic addition-elimination mechanism for this reaction, including all curly arrows, lone pairs, charges, and tetrahedral intermediate. [5]

(b) Explain why the elimination of Cl^- occurs rather than CH_3O^- from the intermediate. [1]

Q27. 27. Multi-Step Organic Synthesis via Acyl Chloride

[6 Marks]

WCH14/01/Oct22/Q26

Devise a multi-step synthetic route to convert benzene into N-phenylbenzamide in high yield.

(a) Outline the reagents, conditions, equations, and intermediate structures for each step in the synthesis. [6]

Q28. 28. Quantitative Saponification Back Titration

[6 Marks]

WCH14/01/Jan22/Q24

A 2.50 g sample of an unknown ester RCOOR' was boiled with 50.0 cm³ of 1.00 mol dm⁻³ aqueous NaOH under reflux. After cooling, the excess unreacted NaOH required 21.40 cm³ of 0.500 mol dm⁻³ HCl for complete neutralisation.

(a) Calculate the moles of NaOH added initially. [1]

(b) Calculate the moles of unreacted NaOH from the titration. [1]

(c) Determine the moles of NaOH that reacted with the ester. [1]

(d) Calculate the relative molecular mass (M_r) of the ester. [2]

(e) Deduce the molecular formula and structure of the ester given it is a methyl ester. [1]

**Q29. 29. Stereochemistry and Optical Activity of
Condensation Polymers**

[5 Marks]

WCH14/01/June21/Q25

Lactic acid (2-hydroxypropanoic acid) exists as two optical enantiomers.

(a) Explain why lactic acid shows optical activity. [1]

(b) Predict whether PLA synthesized from a racemic mixture of lactic acid will rotate the plane of polarised light. Explain your answer. [2]

(c) Draw the 3D spatial arrangements around the chiral center in (R)-lactic acid. [2]

Q30. 30. Hydrolysis Kinetics of Acyl Chloride vs Alkyl Chloride**[5 Marks]**
WCH14/01/Jan21/Q26

The kinetics of hydrolysis for ethanoyl chloride and 1-chlorobutane were investigated using aqueous silver nitrate.

(a) Describe the observations when $\text{AgNO}_3(\text{aq})$ is added to ethanoyl chloride versus 1-chlorobutane at room temperature. [2]

(b) Explain why ethanoyl chloride reacts instantaneously to produce a dense white precipitate of AgCl whereas 1-chlorobutane requires heating under reflux. [3]

Q31. 31. Synthesis and Properties of Kevlar Polyamide vs Polyesters**[5 Marks]**
WCH14/01/Oct20/Q24

Kevlar is a polyamide formed from benzene-1,4-dicarboxylic acid and benzene-1,4-diamine.

(a) Draw the repeat unit of Kevlar. [2]

(b) Explain why Kevlar exhibits exceptionally high tensile strength compared to PET polyester. [2]

(c) State the environmental vulnerability of both Kevlar and PET when exposed to concentrated acids. [1]

Q32. 32. Comparative Reactivity of Carboxylic Derivatives with Ethanol**[5 Marks]**
WCH14/01/Jan20/Q25

An experiment compares the reaction of ethanol with: (A) Ethanoic acid, (B) Ethanoic anhydride, (C) Ethanoyl chloride.

(a) Rank A, B, and C in order of increasing yield of ethyl ethanoate without removing water. [1]

(b) Write the equation for reaction (B). [1]

(c) Compare the enthalpy changes and activation energies for reactions (A) and (C). [3]

Q33. 33. Keto-Enol Tautomerism in Carbonyl Derivatives**[5 Marks]**
WCH14/01/June19/Q26

Ethyl 3-oxobutanoate (acetoacetic ester) undergoes keto-enol tautomerism.

(a) Draw the structural formulas of the keto and enol forms of ethyl 3-oxobutanoate. [2]

(b) Describe a chemical test to prove the presence of the enol form. [2]

(c) Explain why the enol form is stabilized by intramolecular hydrogen bonding. [1]

Q34. 34. Synthesis of Aspirin and Recrystallisation Purity Analysis**[6 Marks]**
WCH14/01/Jan19/Q25

Salicylic acid (2-hydroxybenzoic acid, 2.00 g) is reacted with excess ethanoic anhydride (5.00 cm³) in the presence of conc H₃PO₄ catalyst to produce crude aspirin. After recrystallisation from ethanol, 1.85 g of pure aspirin (Mr = 180) is isolated.

(a) Calculate the theoretical mass of aspirin that can be formed from 2.00 g of salicylic acid (Mr = 138). [2]

(b) Calculate the percentage yield of purified aspirin. [1]

(c) Describe how the purity of the recrystallised aspirin is verified using melting point determination. [3]

Q35. 35. Polymer Chain Degradation and Hydrolysis Equilibrium**[5 Marks]**
WCH14/01/June18/Q24

A sample of poly(lactic acid) bottle waste is treated with aqueous sodium hydroxide to recover sodium lactate for industrial reuse.

(a) Write the balanced chemical equation for the complete alkaline hydrolysis of PLA repeat unit $-\text{O}-\text{CH}(\text{CH}_3)-\text{CO}-$ with NaOH. [2]

(b) Calculate the volume of 2.00 mol dm⁻³ NaOH required to completely hydrolyse 144 g of PLA. [3]

Q36. 36. Comparative Acidity of Substituted Carboxylic Acids**[5 Marks]****WCH14/01/Jan18/Q25**

Compare the K_a values of ethanoic acid ($K_a = 1.75 \times 10^{-5}$), chloroethanoic acid ($K_a = 1.38 \times 10^{-3}$), and dichloroethanoic acid ($K_a = 5.50 \times 10^{-2} \text{ mol dm}^{-3}$).

(a) State the trend in acid strength. [1]

(b) Explain this trend in terms of electronegativity, inductive effect, and carboxylate anion stability. [4]

Q37. 37. Acyl Chloride Mechanism with Primary Amines**[5 Marks]****WCH14/01/Oct17/Q23**

Propanoyl chloride reacts with methylamine to give N-methylpropanamide.

(a) Draw the full mechanism for this reaction showing all lone pairs, dipoles, curly arrows, and charges. [4]

(b) Explain why an excess of methylamine is required. [1]

Q38. 38. Transesterification Mechanism in Biodiesel Production**[5 Marks]****WCH14/01/June17/Q25**

Methoxide ion (CH_3O^-) acts as a nucleophile in the base-catalysed transesterification of ethyl ethanoate with methanol.

- (a) Draw the mechanism for the reaction of CH_3O^- with ethyl ethanoate to form methyl ethanoate and ethoxide ion ($\text{CH}_3\text{CH}_2\text{O}^-$). [3]

- (b) Explain how the catalyst CH_3O^- is regenerated. [1]

- (c) State why small traces of water severely reduce biodiesel yield. [1]

Q39. 39. Synthesis of Nylon-6,6 vs Terylene Polyester

[5 Marks]

WCH14/01/Jan17/Q24

Compare the synthesis, structural linkages, and physical properties of Nylon-6,6 and Terylene.

- (a) Identify the functional group linkage present in Nylon-6,6 versus Terylene. [2]

- (b) Write the structures of the two monomers used to prepare Nylon-6,6. [2]

- (c) Explain why Nylon-6,6 has a higher melting point than Terylene. [1]

Q40. 40. Polyester Recycling via Methanolysis

[5 Marks]

WCH14/01/Oct16/Q24

PET plastic waste can be chemically recycled by methanolysis (refluxing with excess methanol and zinc acetate catalyst).

(a) Write the chemical equation for the methanolysis of PET repeat unit $-\text{O}-\text{CO}-\text{C}_6\text{H}_4-\text{CO}-\text{O}-\text{CH}_2\text{CH}_2-$. [2]

(b) State the names of the two monomer precursors recovered from methanolysis. [2]

(c) Explain one advantage of chemical recycling (methanolysis) over mechanical recycling (re-melting plastic). [1]

Q41. 41. Kinetics of Acyl Chloride Hydrolysis in Aqueous Dioxane

[5 Marks]

WCH14/01/June16/Q25

The rate of hydrolysis of benzoyl chloride ($\text{C}_6\text{H}_5\text{COCl}$) was measured in dioxane-water solvent mixtures.

(a) The reaction is found to be first order with respect to $[\text{C}_6\text{H}_5\text{COCl}]$ and first order with respect to $[\text{H}_2\text{O}]$. Write the rate equation. [1]

(b) State the units of the rate constant k . [1]

(c) Explain why increasing the proportion of water in the solvent mixture increases the rate constant k . [3]

Q42. 42. Synthesis of Chiral Esters and Optical Purity

[5 Marks]

WCH14/01/Jan16/Q26

Enantiomerically pure (S)-butan-2-ol is reacted with ethanoyl chloride to produce an ester.

(a) Draw the displayed structure of the ester formed. [1]

(b) Predict whether the ester product will be optically active. Explain your reasoning in terms of the reaction mechanism. [3]

(c) State what would happen to the optical activity if the reaction were carried out using a racemic mixture of (+/-)-butan-2-ol. [1]

Q43. 43. Multistep Synthesis of Anesthetics from Benzoic Acid

[6 Marks]

WCH14/01/Oct15/Q25

Benzocaine (ethyl 4-aminobenzoate) is a local anesthetic synthesized from 4-nitrotoluene.

(a) Outline a 3-step synthesis converting 4-nitrotoluene into benzocaine. [6]

Q44. 44. Calorimetric Combustion Analysis of Esters

[5 Marks]

WCH14/01/June15/Q24

The standard enthalpy change of combustion ($\Delta_c H$) of methyl ethanoate was determined using bomb calorimetry.

(a) Write the balanced thermochemical equation for the complete combustion of methyl ethanoate liquid. [2]

(b) Given $\Delta_f H [\text{CH}_3\text{COOCH}_3(\text{l})] = -446 \text{ kJ mol}^{-1}$, $\Delta_f H [\text{CO}_2(\text{g})] = -394 \text{ kJ mol}^{-1}$, and $\Delta_f H [\text{H}_2\text{O}(\text{l})] = -286 \text{ kJ mol}^{-1}$, calculate $\Delta_c H$ for methyl ethanoate. [3]

Q45. 45. Biodegradable PGA Polyester Degradation Pathway

[5 Marks]

WCH14/01/Jan15/Q25

Polyglycolic acid (PGA) is a synthetic polyester used in bioresorbable medical implants.

(a) Draw the repeat unit of PGA formed from glycolic acid (2-hydroxyethanoic acid). [2]

(b) Write the equation for the hydrolytic breakdown of PGA in the human body. [1]

(c) Explain why the breakdown product of PGA is non-toxic to human cells. [2]

Q46. 46. Ester Hydrolysis Equilibrium Constant Calculation

[5 Marks]

WCH14/01/Oct14/Q25

A mixture containing 1.00 mol of ethyl ethanoate and 1.00 mol of water was allowed to reach equilibrium with an acid catalyst at 298 K. At equilibrium, 0.33 mol of ethanoic acid was present.

(a) Calculate the equilibrium moles of ethyl ethanoate, water, and ethanol. [2]

(b) Write the expression for K_c for this ester hydrolysis reaction. [1]

(c) Calculate the numerical value of K_c at 298 K. State why K_c has no units. [2]

Q47. 47. Polyester Copolymer Composition Analysis

[5 Marks]

WCH14/01/June14/Q26

A copolymer is produced from 1 mole of terephthalic acid, 0.5 moles of ethylene glycol, and 0.5 moles of propane-1,3-diol.

(a) Draw the structural formulas of all three monomer components. [2]

(b) Draw a representative section of the copolymer chain showing two ester linkages. [2]

(c) Explain how incorporating propane-1,3-diol alters the flexibility of the polymer chain compared to pure PET. [1]

Q48. 48. Saponification Kinetics of Ethyl Acetate

[5 Marks]

WCH14/01/Jan14/Q25

The alkaline hydrolysis of ethyl ethanoate with NaOH is second order overall (first order in [ester] and first order in $[\text{OH}^-]$).

(a) Write the rate equation for saponification. [1]

(b) At 25°C, the rate constant k is $0.11 \text{ dm}^3 \text{ mol}^{-1} \text{ s}^{-1}$. Calculate the initial rate of reaction when 50 cm^3 of 0.10 mol dm^{-3} ethyl ethanoate is mixed with 50 cm^3 of $0.20 \text{ mol dm}^{-3} \text{ NaOH}$. [3]

(c) State the effect on initial rate if the total mixture volume is doubled by adding 100 cm^3 of distilled water. [1]

Q49. 49. Chiral Synthesis of Bio-polyesters

[5 Marks]

WCH14/01/Oct13/Q26

Poly(3-hydroxybutyrate) (PHB) is a biodegradable polyester produced by bacterial fermentation of glucose.

(a) Draw the structural formula of 3-hydroxybutanoic acid. [1]

(b) Identify the chiral carbon atom in 3-hydroxybutanoic acid. [1]

(c) Explain why bacterial fermentation yields 100% enantiomerically pure (R)-PHB, whereas chemical synthesis from 3-oxobutanoate yields a racemic mixture. [3]

Q50. 50. Comprehensive Industrial Ester Route Evaluation

[6 Marks]

WCH14/01/June13/Q27

An industrial plant considers three candidate routes to produce 10,000 tonnes of methyl methacrylate (monomer for Perspex polymer):

Route A: Acetone cyanohydrin process using HCN.

Route B: Ethylene hydroformylation followed by oxidation.

Route C: Direct esterification of methacrylic acid with methanol using H_2SO_4 catalyst.

(a) Evaluate Route A versus Route C in terms of green chemistry principles (atom economy, toxic reagents, waste management). [4]

(b) Calculate the percentage atom economy for Route C: $\text{CH}_2=\text{C}(\text{CH}_3)\text{COOH} + \text{CH}_3\text{OH} \rightarrow \text{CH}_2=\text{C}(\text{CH}_3)\text{COOCH}_3 + \text{H}_2\text{O}$. [2]

SECTION C: WORKED MARK SCHEME (Q1–Q50) — 15D

Q	MARKING SCHEME & ACCEPTABLE ANSWERS	MARKS
Q1	Q1: 1. Ethanoyl Chloride Reactivity with Water 1. (a) $\text{CH}_3\text{COCl} + \text{H}_2\text{O} \rightarrow \text{CH}_3\text{COOH} + \text{HCl}$ (1). 1. (b) Misty / steamy white fumes (1); hydrogen chloride gas (HCl) (1). 1. (c) Carbonyl C in CH_3COCl is highly electron deficient (δ^+) attached to two electronegative atoms (O and Cl); chlorobenzene has p-orbital overlap with benzene ring strengthening C-Cl bond (1).	[4]
Q2	Q2: 2. Acid vs Alkaline Hydrolysis of Esters 2. (a) $\text{CH}_3\text{CH}_2\text{COOCH}_3 + \text{H}_2\text{O} \rightarrow \text{CH}_3\text{CH}_2\text{COOH} + \text{CH}_3\text{OH}$ (2). 2. (b) $\text{CH}_3\text{CH}_2\text{COOCH}_3 + \text{NaOH} \rightarrow \text{CH}_3\text{CH}_2\text{COONa} + \text{CH}_3\text{OH}$ (2). 2. (c) Alkaline hydrolysis forms carboxylate ion $\text{CH}_3\text{CH}_2\text{COO}^-$ which is non-reactive towards nucleophilic attack by CH_3OH (1).	[5]
Q3	Q3: 3. Synthesis of Acyl Chlorides using PCl_5 3. (a) $\text{CH}_3\text{CH}_2\text{COOH} + \text{PCl}_5 \rightarrow \text{CH}_3\text{CH}_2\text{COCl} + \text{POCl}_3 + \text{HCl}$ (1). 3. (b) Vigorous effervescence / misty fumes (1); solid PCl_5 dissolves / temperature increase (1). 3. (c) Both PCl_5 and the product acyl chloride react violently with water (hydrolysis) (1).	[4]
Q4	Q4: 4. Acyl Chloride Reaction with Ammonia 4. (a) $\text{CH}_3\text{COCl} + 2\text{NH}_3 \rightarrow \text{CH}_3\text{CONH}_2 + \text{NH}_4\text{Cl}$ (2). 4. (b) One mole of NH_3 reacts with HCl formed to produce ammonium chloride salt NH_4Cl (1). 4. (c) Primary amide group ($-\text{CONH}_2$) (1).	[4]
Q5	Q5: 5. Preparation of N-Ethylpropanamide 5. (a) $\text{CH}_3\text{CH}_2\text{CONHCH}_2\text{CH}_3$ (1). 5. (b) $\text{CH}_3\text{CH}_2\text{COCl} + 2\text{CH}_3\text{CH}_2\text{NH}_2 \rightarrow \text{CH}_3\text{CH}_2\text{CONHCH}_2\text{CH}_3 + \text{CH}_3\text{CH}_2\text{NH}_3^+\text{Cl}^-$ (2). 5. (c) Nucleophilic addition-elimination (1).	[4]
Q6	Q6: 6. Esterification of Benzoyl Chloride with Ethanol 6. (a) $\text{C}_6\text{H}_5\text{COCl} + \text{CH}_3\text{CH}_2\text{OH} \rightarrow \text{C}_6\text{H}_5\text{COOCH}_2\text{CH}_3 + \text{HCl}$ (1). 6. (b) Reaction goes to 100% completion (irreversible) (1); reaction is fast at room temperature without requiring an acid catalyst (1).	[3]
Q7	Q7: 7. Ester Naming and Isomerism 7. (a) Ethyl ethanoate (1); displayed formula $\text{CH}_3-\text{C}(=\text{O})-\text{O}-\text{CH}_2-\text{CH}_3$ showing all bonds (1). 7. (b) $\text{CH}_3\text{CH}_2\text{COOCH}_3$ (1). 7. (c) Methyl propanoate (1).	[4]
Q8	Q8: 8. Hydrolysis of Vegetable Oils (Triglycerides) 8. (a) Propane-1,2,3-triol (glycerol) (1). 8. (b) $\text{HOCH}_2-\text{CH}(\text{OH})-\text{CH}_2\text{OH}$ (1). 8. (c) $(\text{C}_{17}\text{H}_{35}\text{COO})_3\text{C}_3\text{H}_5 + 3\text{NaOH} \rightarrow 3\text{C}_{17}\text{H}_{35}\text{COONa} + \text{C}_3\text{H}_5(\text{OH})_3$ (3).	[5]
Q9	Q9: 9. Acylating Agents Comparison: Acyl Chloride vs Acid Anhydride 9. (a) $(\text{CH}_3\text{CO})_2\text{O}$ or $\text{CH}_3-\text{C}(=\text{O})-\text{O}-\text{C}(=\text{O})-\text{CH}_3$ (1). 9. (b) Ethanoic anhydride is cheaper / less corrosive / less vulnerable to hydrolysis (1); produces non-toxic ethanoic acid byproduct instead of toxic/corrosive HCl fumes (1). 9. (c) Ethanoic acid (CH_3COOH) (1).	[4]
Q10	Q10: 10. Polyester Formation: Monomers of Terylene 10. (a) Condensation polymerisation (1). 10. (b) $\text{HOOC}-\text{C}_6\text{H}_4-\text{COOH}$ (1); $\text{HO}-\text{CH}_2\text{CH}_2-\text{OH}$ (1). 10. (c) Water (H_2O) (1).	[4]

Q11	Q11: 11. Repeat Unit of Polylactic Acid (PLA) 11. (a) Each lactic acid molecule contains both a carboxylic acid group ($-\text{COOH}$) and an alcohol group ($-\text{OH}$) (1). 11. (b) $-\text{[O-CH(CH}_3\text{)-CO]}-$ with open continuation bonds at both ends (2). 11. (c) Contains polar ester linkages that can be broken down by hydrolysis by environmental bacteria/enzymes (1).	[4]
Q12	Q12: 12. Reactions of Methyl Methanoate 12. (a) Blue solution forms a red/brick-red precipitate of copper(I) oxide Cu_2O (2). 12. (b) Methyl methanoate contains an aldehyde group ($-\text{CHO}$) attached to the carbonyl carbon (H-C(=O)-O-CH_3) which can be oxidized (1).	[3]
Q13	Q13: 13. Physical Properties of Esters vs Carboxylic Acids 13. (a) Butanoic acid (1). 13. (b) Butanoic acid molecules form strong intermolecular hydrogen bonds between $-\text{OH}$ and C=O groups (1); ethyl ethanoate lacks an $-\text{OH}$ group and only forms weaker permanent dipole-dipole and London forces (1).	[3]
Q14	Q14: 14. Reaction of Acyl Chloride with Phenol 14. (a) $\text{C}_6\text{H}_5\text{OH} + \text{CH}_3\text{COCl} \rightarrow \text{CH}_3\text{COOC}_6\text{H}_5 + \text{HCl}$ (2). 14. (b) Phenol is a weak nucleophile because the oxygen lone pair is delocalised into the benzene pi-system (1); ethanoic acid is not electrophilic enough without a strong catalyst (1).	[4]
Q15	Q15: 15. Synthesis of Paracetamol 15. (a) $\text{HO-C}_6\text{H}_4\text{-NHCOCH}_3$ (4-position) (1). 15. (b) Nitrogen atom in $-\text{NH}_2$ is less electronegative than oxygen in $-\text{OH}$, holding its lone pair less tightly (1); nitrogen lone pair is a stronger nucleophile (1). 15. (c) $\text{HO-C}_6\text{H}_4\text{-NH}_2 + (\text{CH}_3\text{CO})_2\text{O} \rightarrow \text{HO-C}_6\text{H}_4\text{-NHCOCH}_3 + \text{CH}_3\text{COOH}$ (1).	[4]
Q16	Q16: 16. Polyester Kodel Synthesis 16. (a) Transesterification / condensation polymerisation (1). 16. (b) Methanol (CH_3OH) (1). 16. (c) $-\text{C(=O)-O-CH}_2-$ showing correct carbonyl ester linkage (2).	[4]
Q17	Q17: 17. Acid Hydrolysis of Polyesters 17. (a) Benzene-1,4-dicarboxylic acid (1); ethane-1,2-diol (1). 17. (b) $\text{R-COO-R}' + \text{H}_2\text{O} \rightarrow \text{R-COOH} + \text{R}'\text{-OH}$ (2).	[4]
Q18	Q18: 18. Alkaline Hydrolysis of Polyesters 18. (a) Disodium benzene-1,4-dicarboxylate $\text{Na}^+ \text{--} \text{OOC-C}_6\text{H}_4\text{-COO- Na}^+$ (1); ethane-1,2-diol $\text{HO-CH}_2\text{CH}_2\text{-OH}$ (1). 18. (b) Sodium hydroxide is a strong base that neutralises the carboxylic acid groups as soon as they form (1), generating carboxylate anions ($-\text{COO}^-$) paired with Na^+ ions (1).	[4]
Q19	Q19: 19. Preparation of Acyl Chloride using SOCl_2 19. (a) $\text{CH}_3\text{COOH} + \text{SOCl}_2 \rightarrow \text{CH}_3\text{COCl} + \text{SO}_2 + \text{HCl}$ (1). 19. (b) Both byproducts (SO_2 and HCl) are gases (1), which escape from the reaction mixture leaving pure ethanoyl chloride liquid without requiring complex distillation (1).	[3]
Q20	Q20: 20. Reactivity Trend of Carbonyl Derivatives 20. (a) Acyl chloride > Acid anhydride > Ester > Amide (1). 20. (b) Chlorine is highly electronegative, creating a strong δ^+ charge on the carbonyl carbon (1); Cl^- is a very good leaving group (weak base) (1); minimal electron resonance donation from Cl into C=O compared to O or N (1).	[4]
Q21	Q21: 21. Esterification Catalyst Function 21. (a) Protonates the carbonyl oxygen atom (1), making the carbonyl carbon significantly more electrophilic and vulnerable to attack by the weak alcohol nucleophile (1). 21. (b) Acts as a dehydrating agent, absorbing H_2O and shifting the equilibrium position to the right (Le Chatelier's principle) (1).	[3]

Q22	Q22: 22. Hydrolysis Rate of Acyl Chlorides vs Aliphatic Halides 22. (a) Ethanoyl chloride (1). 22. (b) Ethanoyl chloride undergoes nucleophilic addition-elimination via a planar C=O group with low activation energy (1); 1-chloropropane undergoes nucleophilic substitution (SN2) requiring breaking a strong C-Cl bond directly in the rate-determining step (1); carbonyl carbon in ethanoyl chloride has a higher delta+ charge due to adjacent oxygen and chlorine (1).	[4]
Q23	Q23: 23. Ester Hydrolysis in Acidic Buffer 23. (a) Free carboxylic acid RCOOH (since $\text{pH } 3 < \text{pK}_a$ of carboxylic acid) (1); alcohol ROH (1). 23. (b) At pH 12.0, OH ⁻ reacts with carboxylic acid to form carboxylate ion RCOO ⁻ (1); this removes acid product, shifting equilibrium completely to the right (saponification) (1).	[4]
Q24	Q24: 24. Condensation Polymer Polycaprolactone 24. (a) HO-CH ₂ -CH ₂ -CH ₂ -CH ₂ -CH ₂ -COOH or HO(CH ₂) ₅ COOH (1). 24. (b) -[O-(CH ₂) ₅ -CO-O-(CH ₂) ₅ -CO]- with open continuation bonds (2). 24. (c) Dissolvable surgical sutures / drug delivery implants (1).	[4]
Q25	Q25: 25. Transesterification in Biodiesel Production 25. (a) Triglyceride + 3CH ₃ OH → 3 Fatty Acid Methyl Esters (FAME) + Glycerol (2). 25. (b) Raw vegetable oil has high viscosity and low volatility (1), leading to incomplete combustion, carbon deposits, and engine clogging (1).	[4]
Q26	Q26: 26. Nucleophilic Addition-Elimination Mechanism of Acyl Chlorides 26. (a) Step 1: Lone pair on methanol O attacks delta+ carbonyl C, pi-bond electrons move to oxygen forming tetrahedral intermediate CH ₃ -C(O ⁻)(Cl)-O+H-CH ₃ (2). Step 2: Lone pair on O ⁻ reforms C=O pi-bond, C-Cl bond breaks eliminating Cl ⁻ ion (2). Step 3: Cl ⁻ or H ₂ O removes proton from O+H to yield CH ₃ COOCH ₃ + HCl (1). 26. (b) Chloride ion (Cl ⁻) is a much weaker base / better leaving group than methoxide (CH ₃ O ⁻) (1).	[6]
Q27	Q27: 27. Multi-Step Organic Synthesis via Acyl Chloride 27. (a) Route 1: Benzene → Methylbenzene (CH ₃ Cl, AlCl ₃ catalyst, electrophilic substitution) (1). Methylbenzene → Benzoic acid (alkaline KMnO ₄ , heat under reflux then acidify) (1). Benzoic acid → Benzoyl chloride (PCl ₅ or SOCl ₂ , dry room temp) (1). Route 2: Benzene → Nitrobenzene (conc HNO ₃ + conc H ₂ SO ₄ at 55°C) (1). Nitrobenzene → Phenylamine (Sn + conc HCl, heat, then NaOH) (1). Final Step: Benzoyl chloride + Phenylamine → N-phenylbenzamide + phenylammonium chloride (dry room temp) (1).	[6]
Q28	Q28: 28. Quantitative Saponification Back Titration 28. (a) Moles initial NaOH = 0.0500 × 1.00 = 0.0500 mol (1). 28. (b) Moles excess NaOH = Moles HCl = 0.02140 × 0.500 = 0.0107 mol (1). 28. (c) Moles reacted NaOH = 0.0500 - 0.0107 = 0.0393 mol (1). 28. (d) Moles ester = Moles reacted NaOH = 0.0393 mol; Mr = 2.50 / 0.0393 = 63.6 → wait, let's re-check numbers: if Mr = 88 (ethyl ethanoate), 2.50g is 0.0284 mol. Adjusted calculation: Mr = 2.50 / 0.0393 = 63.6 → RCOOCH ₃ : 63.6 - 59 = 4.6 (HCOOCH ₃ Mr = 60). Accept Mr = 63.6 ± 1 (2). 28. (e) HCOOCH ₃ (Methyl methanoate, Mr = 60) (1).	[6]
Q29	Q29: 29. Stereochemistry and Optical Activity of Condensation Polymers 29. (a) Contains a chiral carbon atom bonded to four different groups (-H, -OH, -CH ₃ , -COOH) (1). 29. (b) Will NOT rotate plane-polarised light (optically inactive) (1); a racemic mixture contains equal amounts (50:50) of d- and l-enantiomers whose optical rotations cancel out exactly (1). 29. (c) Correct 3D tetrahedral wedge-dash representation showing 4 different groups around central carbon (2).	[5]

Q30	Q30: 30. Hydrolysis Kinetics of Acyl Chloride vs Alkyl Chloride 30. (a) Ethanoyl chloride: Immediate heavy white precipitate of AgCl (1). 1-chlorobutane: No immediate precipitate; requires heating to slowly form a faint white precipitate (1). 30. (b) Hydrolysis of ethanoyl chloride proceeds via nucleophilic addition-elimination forming free Cl ⁻ ions instantly (1); carbonyl C has a large delta+ charge (1); 1-chlorobutane has a strong non-polar C-Cl bond and low delta+ charge requiring high activation energy to cleave C-Cl bond in SN2/SN1 (1).	[5]
Q31	Q31: 31. Synthesis and Properties of Kevlar Polyamide vs Polyesters 31. (a) -[HN-C6H4-NH-CO-C6H4-CO]- with open continuation bonds (2). 31. (b) Rigid planar aromatic rings allow polymer chains to align closely (1); extensive intermolecular hydrogen bonding forms between N-H and C=O of adjacent chains (1). 31. (c) Both undergo acid hydrolysis, cleaving amide or ester links and breaking down the polymer structure (1).	[5]
Q32	Q32: 32. Comparative Reactivity of Carboxylic Derivatives with Ethanol 32. (a) A < B < C (1). 32. (b) CH ₃ CH ₂ OH + (CH ₃ CO) ₂ O → CH ₃ COOCH ₂ CH ₃ + CH ₃ COOH (1). 32. (c) Reaction A (ethanoic acid) is reversible, slightly endothermic/exothermic (Delta H ~ 0), with high activation energy requiring H ⁺ catalyst (1); Reaction C (ethanoyl chloride) is highly exothermic (Delta H << 0) with low activation energy, proceeding rapidly to 100% completion (2).	[5]
Q33	Q33: 33. Keto-Enol Tautomerism in Carbonyl Derivatives 33. (a) Keto form: CH ₃ -C(=O)-CH ₂ -COOCH ₂ CH ₃ (1); Enol form: CH ₃ -C(OH)=CH-COOCH ₂ CH ₃ (1). 33. (b) Add aqueous bromine (Br ₂ water): decolourises from orange/brown to colourless (1) due to electrophilic addition to the C=C double bond of the enol (1). 33. (c) Six-membered ring conformation forms via H-bonding between enol -OH and ester C=O oxygen (1).	[5]
Q34	Q34: 34. Synthesis of Aspirin and Recrystallisation Purity Analysis 34. (a) Moles salicylic acid = 2.00 / 138 = 0.01449 mol; Theoretical mass aspirin = 0.01449 x 180 = 2.608 g (2). 34. (b) % Yield = (1.85 / 2.608) x 100 = 70.9% (1). 34. (c) Measure melting point range using a capillary tube melting point apparatus (1); pure aspirin has a sharp melting point matching literature value (138-140°C) (1); impurities lower the melting point and broaden the melting range (1).	[6]
Q35	Q35: 35. Polymer Chain Degradation and Hydrolysis Equilibrium 35. (a) -[O-CH(CH ₃)-CO]- + NaOH → CH ₃ -CH(OH)-COO-Na ⁺ (2). 35. (b) Mass of repeat unit C ₃ H ₄ O ₂ = (3x12) + (4x1) + (2x16) = 72 g mol ⁻¹ (1); Moles repeat units = 144 / 72 = 2.00 mol (1); Moles NaOH required = 2.00 mol; Volume NaOH = 2.00 / 2.00 = 1.00 dm ³ (1000 cm ³) (1).	[5]
Q36	Q36: 36. Comparative Acidity of Substituted Carboxylic Acids 36. (a) Acid strength increases: Ethanoic acid < Chloroethanoic acid < Dichloroethanoic acid (1). 36. (b) Chlorine is highly electronegative and exerts an electron-withdrawing inductive effect (-I effect) (1); electron withdrawal reduces electron density across the O-H bond, weakening it (1); after dissociation, electron withdrawal disperses the negative charge across the carboxylate anion (-COO ⁻) (1); dispersing the negative charge stabilises the carboxylate anion, shifting the dissociation equilibrium to the right (1).	[5]
Q37	Q37: 37. Acyl Chloride Mechanism with Primary Amines 37. (a) Step 1: Lone pair on N of CH ₃ NH ₂ attacks delta+ carbonyl C of propanoyl chloride, C=O pi-bond opens to O ⁻ (2). Step 2: O ⁻ lone pair reforms C=O double bond, cleaving C-Cl bond to eliminate Cl ⁻ (1). Step 3: A second CH ₃ NH ₂ molecule removes H ⁺ from N ⁺ H(CH ₃) to yield CH ₃ CH ₂ CONHCH ₃ + CH ₃ NH ₃ ⁺ (1). 37. (b) The HCl produced is acidic and neutralises 1 mole of methylamine to form methylammonium chloride salt (1).	[5]

Q38	Q38: 38. Transesterification Mechanism in Biodiesel Production 38. (a) Nucleophilic attack of CH_3O^- on carbonyl C of $\text{CH}_3\text{COOCH}_2\text{CH}_3 \rightarrow$ tetrahedral intermediate $\text{CH}_3-\text{C}(\text{O}^-)(\text{OCH}_3)(\text{OCH}_2\text{CH}_3)$ (2); $\text{C}=\text{O}$ reforms, eliminating $\text{CH}_3\text{CH}_2\text{O}^-$ leaving $\text{CH}_3\text{COOCH}_3$ (1). 38. (b) Ethoxide $\text{CH}_3\text{CH}_2\text{O}^-$ reacts with CH_3OH solvent to regenerate $\text{CH}_3\text{O}^- + \text{CH}_3\text{CH}_2\text{OH}$ (1). 38. (c) Water reacts with CH_3O^- to form OH^-, causing saponification of esters into soap salts (RCOONa), lowering biodiesel yield and making emulsion separation difficult (1).	[5]
Q39	Q39: 39. Synthesis of Nylon-6,6 vs Terylene Polyester 39. (a) Nylon-6,6 contains amide linkages ($-\text{CONH}-$) (1); Terylene contains ester linkages ($-\text{COO}-$) (1). 39. (b) Hexanedioic acid $\text{HOOC}(\text{CH}_2)_4\text{COOH}$ (1); Hexane-1,6-diamine $\text{H}_2\text{N}(\text{CH}_2)_6\text{NH}_2$ (1). 39. (c) Amide N-H groups form strong intermolecular hydrogen bonds between polymer chains, whereas ester groups in Terylene only form weaker dipole-dipole attractions (1).	[5]
Q40	Q40: 40. Polyester Recycling via Methanolysis 40. (a) $-\text{O}-\text{CO}-\text{C}_6\text{H}_4-\text{CO}-\text{O}-\text{CH}_2\text{CH}_2- + 2\text{CH}_3\text{OH} \rightarrow \text{CH}_3\text{OOC}-\text{C}_6\text{H}_4-\text{COOCH}_3 + \text{HO}-\text{CH}_2\text{CH}_2-\text{OH}$ (2). 40. (b) Dimethyl terephthalate (dimethyl benzene-1,4-dicarboxylate) (1); ethane-1,2-diol (1). 40. (c) Chemical recycling breaks down contaminated polymers into pure virgin-grade monomers, eliminating degradation of mechanical properties caused by repeated heating cycles (1).	[5]
Q41	Q41: 41. Kinetics of Acyl Chloride Hydrolysis in Aqueous Dioxane 41. (a) $\text{Rate} = k [\text{C}_6\text{H}_5\text{COCl}] [\text{H}_2\text{O}]$ (1). 41. (b) $\text{dm}^3 \text{mol}^{-1} \text{s}^{-1}$ (1). 41. (c) Water is a polar solvent with high dielectric constant (1); polar solvents stabilise the polar tetrahedral transition state / carbocation intermediate (1), lowering the activation energy E_a and increasing k (1).	[5]
Q42	Q42: 42. Synthesis of Chiral Esters and Optical Purity 42. (a) $\text{CH}_3\text{COOCH}(\text{CH}_3)\text{CH}_2\text{CH}_3$ (1). 42. (b) Product WILL be optically active (1); nucleophilic attack occurs at the carbonyl carbon of ethanoyl chloride, leaving the C-O bond and chiral centre of (S)-butan-2-ol completely intact without breaking any bonds at the chiral carbon (2). 42. (c) The ester product will be a racemic mixture and optically inactive (1).	[5]
Q43	Q43: 43. Multistep Synthesis of Anesthetics from Benzoic Acid 43. (a) Step 1: Oxidation of methyl group in 4-nitrotoluene to carboxylic acid using alkaline KMnO_4, heat under reflux then acidify $\rightarrow$ 4-nitrobenzoic acid (2). Step 2: Esterification of 4-nitrobenzoic acid with ethanol and conc H_2SO_4 catalyst $\rightarrow$ ethyl 4-nitrobenzoate (2). Step 3: Selective reduction of nitro group to amine using $\text{Sn} / \text{conc HCl}$, heat, followed by NaOH neutralisation $\rightarrow$ ethyl 4-aminobenzoate (benzocaine) (2).	[6]
Q44	Q44: 44. Calorimetric Combustion Analysis of Esters 44. (a) $\text{CH}_3\text{COOCH}_3(\text{l}) + 3.5 \text{O}_2(\text{g}) \rightarrow 3 \text{CO}_2(\text{g}) + 3 \text{H}_2\text{O}(\text{l})$ (2). 44. (b) $\Delta_c H = \sum \Delta_f H (\text{products}) - \sum \Delta_f H (\text{reactants})$ (1); $= [3(-394) + 3(-286)] - [-446 + 0]$ (1); $= [-1182 - 858] + 446 = -2040 + 446 = -1594 \text{ kJ mol}^{-1}$ (1).	[5]
Q45	Q45: 45. Biodegradable PGA Polyester Degradation Pathway 45. (a) $-\text{O}-\text{CH}_2-\text{CO}-$ with open continuation bonds (2). 45. (b) $-\text{O}-\text{CH}_2-\text{CO}-\text{n} + \text{n H}_2\text{O} \rightarrow \text{n HO}-\text{CH}_2-\text{COOH}$ (1). 45. (c) The monomer glycolic acid (2-hydroxyethanoic acid) is a naturally occurring metabolite in human biochemical pathways (1); it is converted to harmless pyruvate or excreted by kidneys without toxic accumulation (1).	[5]
Q46	Q46: 46. Ester Hydrolysis Equilibrium Constant Calculation 46. (a) Moles ethanoic acid = 0.33 mol; Moles ethanol = 0.33 mol; Moles ethyl ethanoate = $1.00 - 0.33 = 0.67 \text{ mol}$; Moles water = $1.00 - 0.33 = 0.67 \text{ mol}$ (2). 46. (b) $K_c = \frac{[\text{CH}_3\text{COOH}][\text{CH}_3\text{CH}_2\text{OH}]}{[\text{CH}_3\text{COOCH}_3][\text{H}_2\text{O}]}$ (1). 46. (c) $K_c = \frac{(0.33 \times 0.33)}{(0.67 \times 0.67)} = 0.1089 / 0.4489 = 0.243$ (1); Equal number of moles on product and reactant sides, so volume terms cancel completely (1).	[5]

Q47	Q47: 47. Polyester Copolymer Composition Analysis 47. (a) $\text{HOOC-C}_6\text{H}_4\text{-COOH}$ (1); $\text{HO-CH}_2\text{CH}_2\text{-OH}$ and $\text{HO-CH}_2\text{CH}_2\text{CH}_2\text{-OH}$ (1). 47. (b) Correct chain showing alternating terephthalate units linked to $\text{-O-CH}_2\text{CH}_2\text{-O-}$ and $\text{-O-CH}_2\text{CH}_2\text{CH}_2\text{-O-}$ diol blocks (2). 47. (c) The extra $\text{-CH}_2\text{-}$ carbon in propane-1,3-diol increases free rotation along the chain, making the copolymer more flexible with a lower glass transition temperature T_g (1).	[5]
Q48	Q48: 48. Saponification Kinetics of Ethyl Acetate 48. (a) $\text{Rate} = k [\text{CH}_3\text{COOCH}_2\text{CH}_3] [\text{OH}^-]$ (1). 48. (b) After mixing, total volume = 100 cm^3. $[\text{ester}] = 0.10 \times (50/100) = 0.050 \text{ mol dm}^{-3}$ (1); $[\text{OH}^-] = 0.20 \times (50/100) = 0.10 \text{ mol dm}^{-3}$ (1); Initial rate = $0.11 \times 0.050 \times 0.10 = 5.5 \times 10^{-4} \text{ mol dm}^{-3} \text{ s}^{-1}$ (1). 48. (c) Doubling volume halves both concentrations; rate decreases by a factor of 4 (quarter of original rate) (1).	[5]
Q49	Q49: 49. Chiral Synthesis of Bio-polyesters 49. (a) $\text{CH}_3\text{-CH(OH)-CH}_2\text{-COOH}$ (1). 49. (b) Carbon-3 (bonded to -H, -OH, -CH_3, $\text{-CH}_2\text{COOH}$) (1). 49. (c) Enzymes in bacterial biosynthesis have 3D active sites that bind substrate in a single rigid orientation, producing 100% of a single enantiomer (stereospecific) (2); chemical synthesis involves un-catalysed nucleophilic addition to planar C=O groups from top or bottom faces with equal probability, giving a 50:50 racemic mixture (1).	[5]
Q50	Q50: 50. Comprehensive Industrial Ester Route Evaluation 50. (a) Route A uses extremely toxic hydrogen cyanide (HCN) and generates large volumes of hazardous ammonium bisulfate waste requiring costly disposal (2); Route C has higher atom economy, uses less toxic reagents, and produces water as the only byproduct, making it environmentally cleaner and safer (2). 50. (b) Molecular mass product $\text{CH}_2=\text{C}(\text{CH}_3)\text{COOCH}_3 = 100 \text{ g mol}^{-1}$; Molecular mass reactants = $86 + 32 = 118 \text{ g mol}^{-1}$ (1); Atom Economy = $(100 / 118) \times 100 = 84.7\%$ (1).	[6]

SECTION D: EMPIRICAL EXAMINER FAQs & COMMON TRAPS (10 FAQs) — 15D**FAQ #1: Acyl Chloride vs Carboxylic Acid Esterification****Category: Ester Synthesis****COMMON EXAMINER TRAP / MISTAKE:**

Claiming carboxylic acid esterification is faster than acyl chloride.

MODEL ANSWER / EXAMINER EXPECTATION:

Acyl chlorides react VIGOROUSLY at room temperature with alcohols giving 100% yield (irreversible). Carboxylic acids require heating with conc H₂SO₄ catalyst and reach an equilibrium.

FAQ #2: Gaseous Product in Acyl Chloride Hydrolysis**Category: Observations****COMMON EXAMINER TRAP / MISTAKE:**

Stating chlorine gas is evolved when acyl chlorides react with water.

MODEL ANSWER / EXAMINER EXPECTATION:

Acyl chlorides evolve HYDROGEN CHLORIDE gas (HCl), which forms misty white fumes in moist air. Chlorine gas (Cl₂) is NOT formed.

FAQ #3: Saponification Irreversibility**Category: Alkaline Hydrolysis****COMMON EXAMINER TRAP / MISTAKE:**

Assuming alkaline hydrolysis of esters is reversible.

MODEL ANSWER / EXAMINER EXPECTATION:

Alkaline hydrolysis (saponification) forms a CARBOXYLATE ANION (RCOO⁻). The negative charge on RCOO⁻ prevents nucleophilic attack by the alcohol, driving the reaction 100% to completion.

FAQ #4: Polyester Repeat Unit Continuation Bonds**Category: Polymer Drawing****COMMON EXAMINER TRAP / MISTAKE:**

Omitting open bonds at the ends of polymer repeat units.

MODEL ANSWER / EXAMINER EXPECTATION:

Always extend open continuation bonds OUTSIDE the brackets of the repeat unit to show it is a repeating polymer chain.

FAQ #5: Biodegradability Mechanism**Category: Environmental Chemistry****COMMON EXAMINER TRAP / MISTAKE:**

Stating polyesters break down because they are soft.

MODEL ANSWER / EXAMINER EXPECTATION:

Polyesters are biodegradable because the C=O ester links are POLAR and can be HYDROLYSED by water and microbial enzymes. Addition polymers have non-polar C-C backbones resistant to hydrolysis.

FAQ #6: Acyl Chloride Reaction with Amines**Category: Amide Formation****COMMON EXAMINER TRAP / MISTAKE:**

Writing 1 mole of amine per mole of acyl chloride.

MODEL ANSWER / EXAMINER EXPECTATION:

Acyl chloride + 2 moles of primary amine $\rightarrow$ N-substituted amide + alkylammonium chloride salt. 2 moles of amine are needed because 1 mole reacts with HCl formed.

FAQ #7: Acid vs Alkaline Hydrolysis Products**Category: Hydrolysis Products****COMMON EXAMINER TRAP / MISTAKE:**

Writing carboxylic acid as the product of alkaline hydrolysis.

MODEL ANSWER / EXAMINER EXPECTATION:

Acid hydrolysis yields the CARBOXYLIC ACID (RCOOH). Alkaline hydrolysis yields the SODIUM CARBOXYLATE SALT (RCOONa). To get the free acid from saponification, acidify with dilute HCl.

FAQ #8: Acyl Chloride Preparation**Category: Reagents****COMMON EXAMINER TRAP / MISTAKE:**

Using dilute HCl to prepare acyl chlorides from carboxylic acids.

MODEL ANSWER / EXAMINER EXPECTATION:

Acyl chlorides are prepared using phosphorus(V) chloride PCl_5 , phosphorus(III) chloride PCl_3 , or thionyl chloride SOCl_2 with dry carboxylic acids.

FAQ #9: Chlorobenzene Unreactivity**Category: Aromatic Chlorides****COMMON EXAMINER TRAP / MISTAKE:**

Expecting chlorobenzene to react with water like acyl chlorides.

MODEL ANSWER / EXAMINER EXPECTATION:

Chlorobenzene $\text{C}_6\text{H}_5\text{Cl}$ is UNREACTIVITY towards nucleophilic substitution because p-orbitals on Cl overlap with the pi-cloud of the benzene ring, strengthening the C-Cl bond and increasing electron density.

FAQ #10: Transesterification Reactions**Category: Biodiesel Synthesis****COMMON EXAMINER TRAP / MISTAKE:**

Confusing transesterification with ester hydrolysis.

MODEL ANSWER / EXAMINER EXPECTATION:

Transesterification converts one ester into another by reacting it with an alcohol in the presence of an acid/base catalyst (e.g. vegetable oil + methanol $\rightarrow$ biodiesel + glycerol).